Edge-On Geometry Raises a Public Orbital Data-Center Radiator Model's Coded Equilibrium Temperature by 6.35 K

An Exact Earth View-Factor Correction, Its Decomposition, and a Verified Orbit-Coupled Transient Extension

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Abstract

A prominent publicly accessible first-principles model of orbital data-center radiator behavior is Andrew McCalip's open "Space Datacenters" model, which estimates a sun-tracking bifacial panel's equilibrium temperature from a closed heat balance with an approximate Earth view factor: a cosine-of-tilt heuristic with a lower floor at 5% of the nadir value. We reproduce that model independently to floating-point roundoff and then change only the view-factor approximation, holding every other model choice and constant fixed. At the model's own default geometry (orbit beta angle $\beta = 90°$, 550 km) the sun-tracking panel is *edge-on* to Earth: its normal tracks the Sun, which at $\beta = 90°$ is normal to the orbit plane, while nadir lies in the plane 90° away. The exact per-face edge-on Earth view factor there is 0.25777. The public code, however, produces an orbit-averaged per-face value of 0.02118, even though its nominal 5%-of-nadir floor is 0.04237; the additional halving arises from the floating-point treatment of $\cos 90°$ in the branch logic. Replacing the coded value with the exact view factor raises the coded equilibrium from 335.75 K to 342.10 K, a +6.35 K correction to the public code *as executed*. We decompose this: relative to the intended 5% floor the geometric correction is +5.77 K, with the remaining +0.58 K an edge-case implementation artifact. The correction is positive at every sampled β and increases monotonically on a 0.5° grid, with its largest sampled value at the default. We then extend the static balance to an orbit-coupled, time-dependent model—a one-node radiator with thermal mass driven by an orbit-varying effective sink—and quantify the averaging-load bias: at periodic steady state energy balance forces $\langle T^4 \rangle = T^4_{\text{steady}}$, so the arithmetic-mean temperature sits at or below the steady value, while any nonzero periodic ripple necessarily has a peak above it; the operational penalty of a steady-state sizing is peak under-prediction, not a mean offset. The headline view-factor decomposition, equilibrium temperatures, beta table, and transient-bias examples are reproduced by a paper-specific verification script pinned to the package release; the McCalip replication is separately locked against a SHA-256-pinned Node oracle by the package test suite. We are explicit about scope: this is cross-model verification (our exact geometry corrects his coded geometry), not validation against a real radiator, for which we identified no public flight data for this configuration.

1 Introduction

Two companion papers established, within a gray-body effective-sink radiator model, the exact thermodynamic bounds that govern heat rejection for an orbital data center [1] and applied those

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bounds to the first announced industrial design point, SpaceX's AI1 satellite [2, 3, 4]. Both are static. Neither addresses how the radiative environment varies as the panel moves around its orbit, nor how a panel with thermal mass responds to that variation in time. This paper takes up both, and in doing so produces a concrete, verifiable correction to an existing public model.

A prominent publicly accessible first-principles model of orbital-datacenter radiator behavior is Andrew McCalip's open "Space Datacenters" model [5]. It is an unusually transparent artifact: a closed heat balance for a sun-tracking bifacial panel, with all constants and geometry stated in source. Like any reduced-order model it makes many simplifications—a gray, isothermal bifacial balance; constant surface properties; fixed Earth-IR and albedo parameters; a simplified albedo phase scaling; a 72-point orbit average; a uniform spherical Earth; and no detailed conduction, plumbing, or fin efficiency. We do not survey those. We change only one element, the Earth view factor, holding every other model choice and constant fixed, and quantify the consequence.

Our contribution is in three parts. First (Section 2) we give the exact tilted-plate-to-sphere Earth view factor, including a closed form at the edge-on geometry, and contrast it with the heuristic. Second (Section 3, the headline result) we reproduce McCalip's model independently and substitute the exact view factor into his own heat balance, isolating a +6.35 K correction at his default $\beta = 90°$ edge-on geometry, decomposing it into a model-form geometric part and an implementation-edge-case part, and tabulating the correction across β. Third (Sections 4–5) we extend the static balance to an orbit-coupled effective sink and a one-node transient solver, and quantify the averaging-load bias between a time-dependent solution and a steady-state sizing.

Everything quantitative here is reproduced by an open Python implementation and a paper-specific verification script (Section 7). The replication of McCalip's model is locked against a SHA-256-pinned regeneration of his original JavaScript at a specific commit. The software was developed and hardened through an author-directed adversarial review process with an AI system performing independent re-derivation and computer-algebra checks; we treat that as a software-review and reproducibility record (Section 6), not as external scholarly peer review, and we are correspondingly careful about the status of the claims: this is cross-model verification—our exact geometry corrects his coded geometry—and not validation against measured radiator performance.

2 The exact Earth view factor and the heuristic it replaces

2.1 Effective-sink model

We use the same gray-body effective-sink radiator model as the companion papers [1]. A gray, diffuse, isothermal surface of emissivity $\varepsilon \in (0, 1]$ at temperature T exchanges with a lumped environment summarized by an effective sink temperature T_s^{eff}, so the net rejected flux is $q = \varepsilon\sigma\,(T^4 - (T_s^{\mathrm{eff}})^4)$. The environment seen by an orbiting panel is not constant: Earth infrared and reflected-solar (albedo) loads enter through a geometric Earth view factor that depends on the angle between the panel face normal and nadir, and on orbital phase. Figure 1 shows a representative effective-sink profile around one orbit; the swing in T_s^{eff} is what makes the view factor a first-order modeling choice rather than a detail.

2.2 Exact tilted-plate-to-sphere view factor

Treat Earth as a uniform Lambertian sphere subtending angular radius $\theta = \arcsin(R_\oplus/r)$ at orbital radius r, centered on nadir, and let γ be the angle between the plate normal and nadir. For a differential planar element—an effectively exact approximation for a spacecraft panel, whose dimensions are negligible relative to r—the configuration factor is the cosine-weighted solid angle

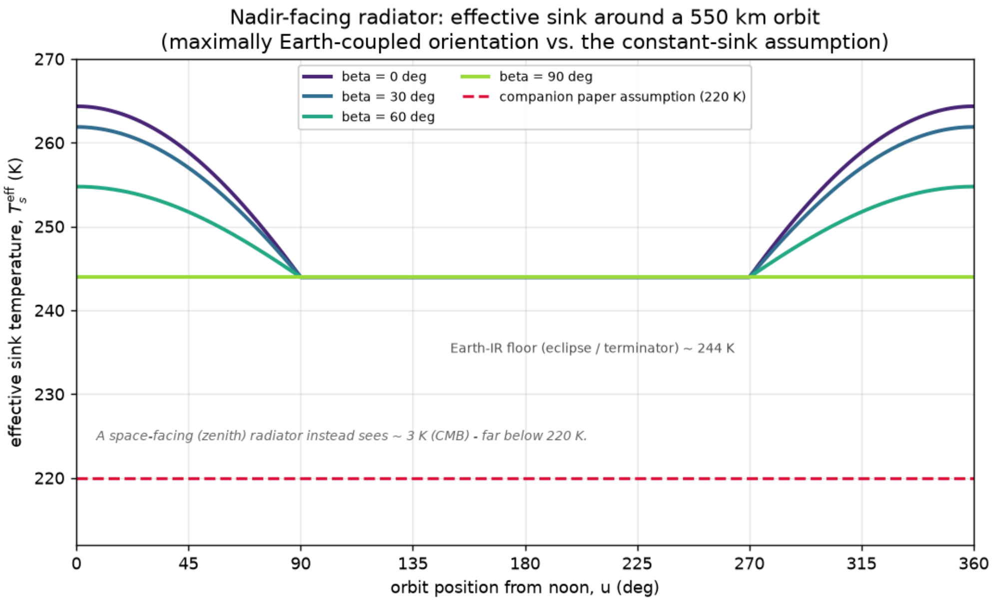

Figure 1: Illustrative effective-sink temperature T_s^{eff} around a 550 km orbit for a *nadir-facing* radiator at several orbit beta angles. For this fixed nadir attitude the Earth view factor is constant over the orbit; the phase variation shown is driven by the subpoint-albedo term (Section 4), and vanishes toward $\beta = 90°$. The figure illustrates that the radiated-against environment breathes with orbital position; the orientation-dependent view-factor effect that drives the correction of Section 3 is a separate, larger effect for a sun-tracking panel.

over the part of the disk above the plate's horizon [6]. It has the closed limiting forms

$$F(\gamma) = \begin{cases} \cos\gamma\, \sin^2\theta, & \gamma \le \frac{\pi}{2} - \theta \quad \text{(Earth fully above the horizon)}, \\ 0, & \gamma \ge \frac{\pi}{2} + \theta \quad \text{(Earth fully set)}, \end{cases} \tag{1}$$

and for the intermediate regime $\frac{\pi}{2} - \theta < \gamma < \frac{\pi}{2} + \theta$ the horizon clips the disk. The edge-on case $\gamma = \frac{\pi}{2}$—the paper's headline geometry—admits the closed form

$$F\left(\tfrac{\pi}{2}\right) = \frac{\theta - \sin\theta\cos\theta}{\pi}, \tag{2}$$

obtained by integrating the cosine-weighted solid angle over the half of the disk above the horizon. At 550 km, $\theta = \arcsin(6371/6921)$ and (2) gives $F(\frac{\pi}{2}) = 0.257773$ per face. The package computes the general intermediate regime by a fixed 48-node piecewise Gauss–Legendre rule, split at the horizon crossings, with a demonstrated absolute error of order 10^{-9} against independent checks; at edge-on it reproduces (2) to $< 10^{-6}$, so the headline value is independently checkable from (2) without the software.

McCalip's model instead approximates F by a cosine-of-tilt expression with a lower floor at 5% of the nadir value: per face, $F_{\mathrm{heur}}(\gamma) = F_{\mathrm{nadir}} \cos\gamma$ for $\cos\gamma > 0$ and $F_{\mathrm{nadir}} \cdot 0.05$ otherwise, with $F_{\mathrm{nadir}} = \sin^2\theta$. Near nadir this agrees with (1) to leading order. Near the horizon the true factor approaches a small but nonzero grazing value while the heuristic collapses onto its arbitrary floor.

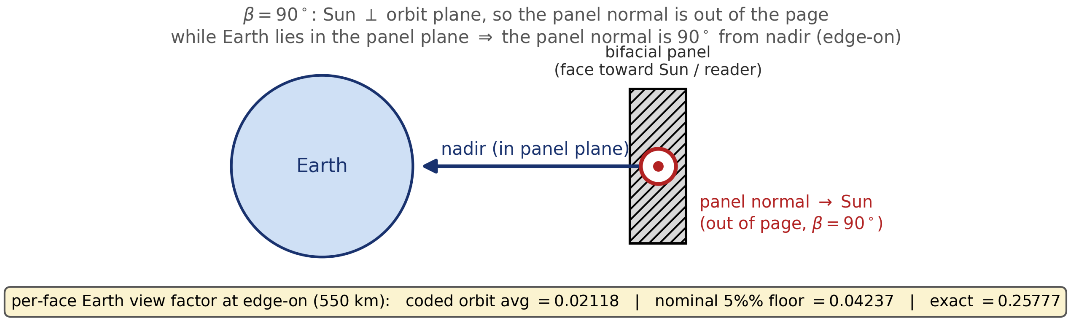

Figure 2: Why the default geometry is edge-on. At $\beta = 90°$ the Sun is perpendicular to the orbit plane, so a sun-tracking panel's normal points out of the plane while Earth (nadir) lies within it—the panel normal is 90° from nadir. The three per-face Earth view factors in the callout are computed from the version-pinned package: the actual coded orbit average (0.02118), the nominal 5% floor (0.04237), and the exact edge-on value (0.25777).

3 Replication and the edge-on correction

3.1 Faithful replication

An independent Python port of McCalip's `static/js/math.js`—using his exact constants, including his truncated $\sigma = 5.67 \times 10^{-8}$ and rounded deep-space temperature $T_{\text{space}} = 3$ K—matches a frozen Node regeneration of his original code (pinned commit `d1e4238`) to a maximum relative error of $\sim 4 \times 10^{-14}$ across all 297 compared values in eleven parameter cases: floating-point roundoff. The model is replicated, not approximated, and the replication target is pinned so it cannot drift.

3.2 The default geometry is edge-on

McCalip's default is a sun-tracking bifacial panel at $\beta = 90°$ and 550 km. At $\beta = 90°$ the Sun is normal to the orbit plane; a sun-tracking panel therefore holds its face normal out of the orbit plane, while nadir lies in the plane, so the panel normal is 90° from nadir (Figure 2). Each face sees Earth edge-on for the entire orbit—the worst case for the cosine-of-tilt heuristic, and the default.

Three per-face view factors must be distinguished at the edge-on default (Table 1). The model's *intended* floor is 5% of nadir, $0.05 \times 0.84738 = 0.04237$. Its *actual* orbit-averaged output is 0.02118: at $\beta = 90°$ the code evaluates $\cos(90°)$ as $\approx 6.12 \times 10^{-17}$ rather than exact zero, so in the 72-point orbit loop roughly half the samples take the tiny-positive-cosine branch and return a value near zero while the other half take the nonpositive branch and return the floor; the average is almost exactly half the floor. The *exact* edge-on value is 0.25777. Thus the exact factor exceeds the coded output by 12.2× and the intended floor by 6.1×.

3.3 The correction and its decomposition

Substituting the exact per-face view factor into McCalip's own heat balance, and changing nothing else, moves his default coded equilibrium temperature

$$\begin{gathered} 335.749538 \text{ K (coded, replicated)} \longrightarrow 342.099222 \text{ K (exact edge-on VF)}, \\ \Delta T = +6.349684 \text{ K}. \end{gathered} \tag{3}$$

Table 1: Per-face Earth view factor at the edge-on default ($\beta = 90°$, 550 km), and the equilibrium-temperature obtained by feeding each value (both faces) through McCalip's own heat balance, changing nothing else.

Per-face Earth view factor	value	heat-balance T_{eq} (K)
Actual coded $\beta = 90°$ orbit average	0.02118447	335.749538
Nominal intended 5% floor	0.04236893	336.332909
Exact edge-on view factor (2)	0.25777283	342.099222

Table 2: Decomposition of the headline correction at the edge-on default. The +6.35 K is the correction to the actual coded output; it separates into a model-form geometric correction (intended floor → exact) and an implementation edge-case artifact.

Comparison	ΔT (K)
Actual coded output → exact geometry	+6.349684
of which: nominal 5% floor → exact (model-form geometry)	+5.766313
of which: branch artifact (coded → nominal floor)	+0.583371

This is the correction to the public code *as executed*. It decomposes cleanly (Table 2): replacing the intended 5% floor by the exact geometry accounts for +5.766313 K (a model-form geometric error), and the remaining +0.583371 K is the floating-point branch artifact in the orbit average described above (a numerical implementation effect, not a geometric-model error). A self-consistency check confirms the attribution: feeding McCalip's own view factors back through the same heat-balance code reproduces his number to roundoff, so the shift is due to the view factor and not to any incidental difference in the balance.

The correction is positive at every sampled orbit beta angle, because a sun-tracking panel's faces, averaged over the orbit, couple to Earth more strongly than the heuristic reports; across the evaluated range $\beta \in [0°, 90°]$ the exact orbit-averaged bifacial Earth coupling exceeds the heuristic result, so the temperature correction stays positive at every sampled β, and increases monotonically on a dense grid (step 0.5° over $[0°, 90°]$; minimum sampled increment $+1.06 \times 10^{-4}$ K), beyond the displayed grid of Table 3 and Figure 3. We claim monotonicity at the sampled points; continuous monotonicity between samples is not separately proved.

Remark 1. The sign is intuitive. Underestimating the Earth view factor underestimates absorbed Earth IR and albedo, understating the environmental load and hence the temperature the panel must reach to reject it; correcting the view factor upward raises the predicted temperature.

4 Orbit-coupled effective sink

The static correction fixes one point; a flight panel sweeps a range. We model the effective sink around the orbit as $T_s^{\mathrm{eff}}(u)$, with orbital phase $u \in [0°, 360°)$, combining the phase-independent Earth-IR term and a reflected-solar (albedo) term driven by a subpoint-albedo factor $\max(0, \cos\beta \cos u)$, both weighted by the exact view factor of Section 2; environmental flux conventions follow standard references [9]. The orbit mean of the albedo factor has the closed form $\langle \max(0, \cos\beta \cos u) \rangle = \cos\beta/\pi$ for $\beta \in [0°, 90°]$, a grid-free reference for the environmental drive used in the convergence checks of Section 5.

We flag one model-form limitation explicitly. The reflected-solar term is a *subpoint* approximation:

Table 3: Equilibrium-temperature correction versus orbit beta angle, from substituting the exact per-face Earth view factor into McCalip's own heat balance (i.e. relative to his coded output at each β). The six-decimal values elsewhere in this paper are computational fingerprints for reproducibility, not claims of physical accuracy.

β (deg)	coded T_{eq} (K)	exact-VF T_{eq} (K)	correction (K)
0	349.58	351.53	+1.94
15	348.94	350.98	+2.04
30	347.12	349.53	+2.41
45	344.42	347.63	+3.22
60	341.28	345.66	+4.38
75	338.24	343.79	+5.55
90	335.75	342.10	+6.35

it samples reflectance only at the sub-satellite point, so it nulls albedo whenever the subpoint is dark and, at $\beta = 90°$, reports zero orbit-averaged albedo and a phase-independent sink. A disk-integrated model accounting for the sunlit Earth crescent within the radiator's field of view would not vanish there. The subpoint factor is retained as a documented approximation for the relative, orbit-averaged behavior we report, not asserted as physics; a disk-integrated replacement is left to future work.

5 Transient extension and the averaging-load bias

5.1 One-node model

A real radiator panel has thermal mass and cannot follow the orbit-varying sink instantaneously; it lags and ripples. We model this with a one-node energy balance per unit area,

$$C\,\frac{dT}{dt} = q_{\mathrm{load}} - \varepsilon\sigma\big(T^4 - T_s^{\mathrm{eff}}(t)^4\big), \tag{4}$$

with C the areal heat capacity ($\mathrm{J\,m^{-2}\,K^{-1}}$) and q_{load} the internal waste-heat flux. Equation (4) is integrated with fixed-step RK4 and marched to a periodic steady state. Figure 4 shows the converged temperature history.

5.2 The averaging-load bias

Does sizing a radiator at the orbit-averaged sink mis-estimate its behavior? At periodic steady state, integrating (4) over one orbit makes the stored-energy change vanish, so energy balance forces the fourth-power mean to equal the steady value exactly,

$$\langle T^4\rangle = \frac{q_{\mathrm{load}}}{\varepsilon\sigma} + \langle (T_s^{\mathrm{eff}})^4\rangle = T_{\mathrm{steady}}^4, \tag{5}$$

with T_{steady} the steady solution at the fourth-power-averaged sink. Because $x^{1/4}$ is concave, the power-mean inequality gives $\langle T\rangle \le \langle T^4\rangle^{1/4} = T_{\mathrm{steady}}$: the arithmetic-mean temperature sits at or below the steady value. The steady, averaged-load sizing therefore does not under-predict the mean. What it misses is the *peak*: any nonzero periodic ripple necessarily carries the panel above T_{steady} for part of each orbit (if the waveform is constant—as the subpoint model becomes at $\beta = 90°$—the peak, mean, and steady values coincide and there is no excess). The peak excess is the engineering-relevant penalty.

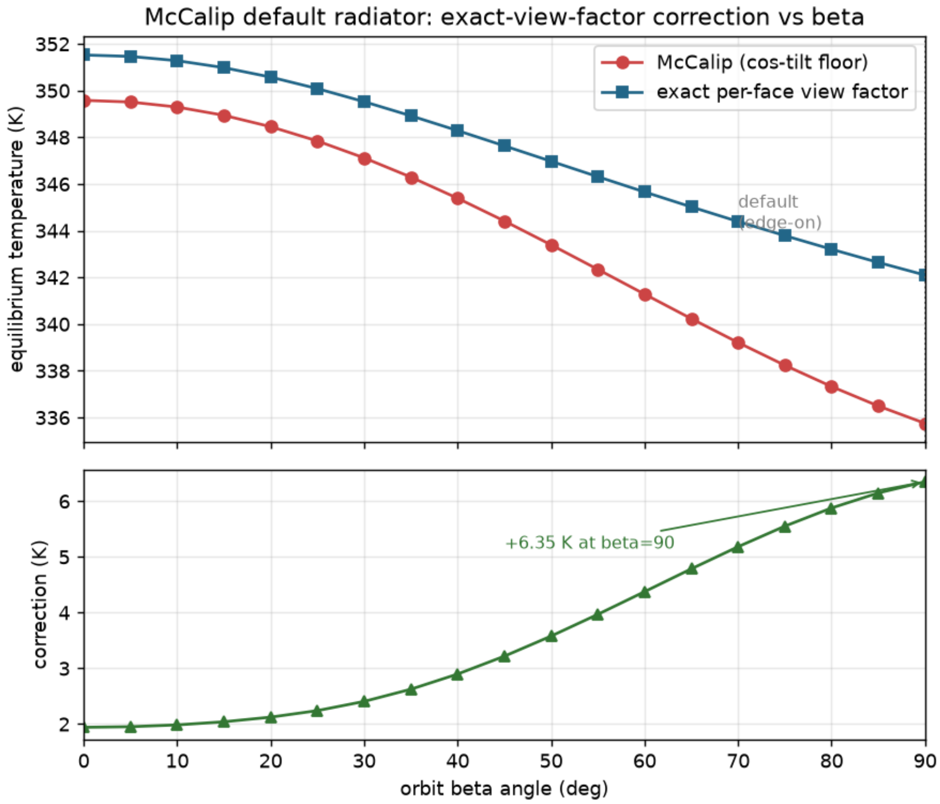

Figure 3: On the evaluated 0.5° beta grid, the equilibrium-temperature correction grows monotonically toward the edge-on default. As β increases, the sun-tracking panel spends the orbit closer to the edge-on regime, where the heuristic's orbit-averaged error is largest—so the implied temperature error is largest exactly where the model is operated.

Table 4 reports both quantities at a representative operating point across a range of areal heat capacity. The signed mean bias is small and nonpositive, as (5) requires; the peak excess is positive and grows as thermal mass falls, from 0.8 K for a heavy panel to 4.5 K for a light one. The identity (5) is verified numerically to the solver tolerance at every case.

5.3 Convergence certificate

Because the periodic steady state and its peak are the load-bearing quantities, the solver returns a result only when it is certified along three independent axes: periodic closure (start-to-end orbit temperature change below tolerance); a scale-aware energy balance (the orbit-mean net flux, as an equivalent temperature error, below tolerance, which catches a high-thermal-inertia trap that closure alone misses); and temporal resolution (independently converged solutions at N, $2N$, and $4N$ steps per orbit are compared by a grid-free analytic forcing-quadrature check, a direct $N \to 4N$ comparison, and a pointwise waveform comparison on a common phase grid, with a conservative safety margin). The temporal residual is reported as a refinement-based error estimate rather than a guaranteed bound, and peak-phase drift is reported separately because an amplitude bound does not bound the time of the peak. These checks are what let the numbers in Table 4 be quoted without hidden under-resolution.

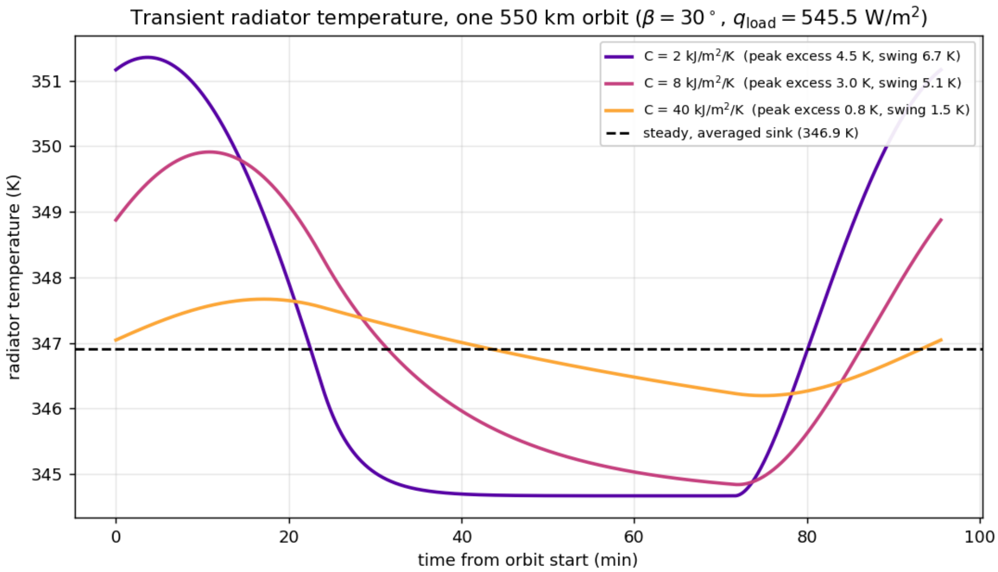

Figure 4: One-node panel temperature over a converged orbit at periodic steady state (550 km, $\beta = 30°$, $q_{\text{load}} = 545.5\ \text{W m}^{-2}$, the same point as Table 4), for three areal heat capacities. Thermal mass turns the sink swing into a damped, phase-lagged ripple; the peak excess over the steady-state sizing temperature, not the mean, is the operationally relevant quantity.

Table 4: Averaging-load bias at 550 km, $\beta = 30°$, $q_{\text{load}} = 545.5\ \text{W m}^{-2}$. "Mean bias" is $\langle T \rangle - T_{\text{steady}}$ (nonpositive, per (5)); "peak excess" is $T_{\max} - T_{\text{steady}}$. τ/P is the radiative time constant over the orbital period.

C ($\text{J m}^{-2}\text{K}^{-1}$)	mean bias (K)	peak excess (K)	swing (K)	τ/P
2000	−0.028	4.45	6.69	0.04
8000	−0.014	3.01	5.07	0.16
40000	−0.001	0.77	1.47	0.81

6 Verification, reproducibility, and scope

The claims here occupy specific, deliberately limited levels of a verification-and-validation hierarchy [7]. *Mathematical* verification (the equations follow from the assumptions) and *software* verification (the code computes them) are covered by the package's assertion suite and by the paper-specific verification script of Section 7. *Cross-model* verification (independent implementations of the *same* model agree) is covered two ways: our replication of McCalip's model against his SHA-256-pinned, semantically regenerated original, and our exact view factor checked both against an independent integrator and against the closed form (2). The headline result of Section 3 is a cross-model verification statement: our exact geometry corrects his coded geometry within his own balance.

We do *not* claim *model-form* validation (assumptions match physical benchmarks) or *system* validation (predictions match hardware or flight data). The subpoint-albedo approximation and the sun-shielded attitude assumption are model-form gaps, held open and labeled; closing them would require the kind of detailed environmental heat-load and view-factor modeling described in standard spacecraft thermal-analysis practice [8]. We identified no publicly available flight data for this specific orbital-data-center radiator configuration as of June 2026, and we assert no validation against any.

On methodology: the implementation was developed and hardened through an author-directed adversarial review process in which an AI system (GPT-5.5 with a computer-algebra plugin) performed independent re-derivation, counterexample search, and symbolic checks across successive revisions, ending without a remaining flagged defect. That process is a software-review and reproducibility record—the review reports are archived as software-review artifacts—and not external scholarly peer review or evidence of physical validation; the reviewer was an AI system operating under the author's direction, not an independent institution or journal referee.

7 Code and data availability

The model, tests, frozen oracle, figures, and a paper-specific verification script are openly available in the project repository `github.com/dan-lee-odinson/orbital-thermal-bounds` at release `v1.0.0`, immutable commit `322fc44db8dc175450ac2e9eb918fe3a1758b2b1` (a Zenodo software DOI will be minted at archival). The package targets Python ≥ 3.10 with `numpy` (and `CoolProp==7.2.0` for the optional coolant screen); the McCalip oracle is regenerated with Node (≥ 18). To reproduce:

```
pip install -e ".[fluids]"
python verify_paper3.py   # decomposition, Tables 1-4, Eq. (5)
pytest                    # full suite + SHA-256 oracle freeze
```

`verify_paper3.py` is pinned to `orbital-thermal==1.0.0` and recomputes the view-factor decomposition (Table 1), the three balances and decomposition (Table 2), the beta-correction table (Table 3), the dense-grid monotonicity increment, and the periodic identity (5) with the full Table 4 rows; the McCalip replication against the SHA-256-pinned Node oracle is locked separately by the package suite (`tests/test_mccalip_replication.py`, oracle pinned at upstream commit `d1e4238d3d3f4924e5ca65bafbd4ba5b39af2eb8`, MIT). Each figure and table is regenerated by a named script: Figure 1 by `scripts/plot_effective_sink.py`, Figure 2 by `scripts/plot_edge_on_geometry.py`, Figure 3 by `scripts/plot_mccalip_correction.py`, and Figure 4 by `scripts/plot_transient.py` (at $\beta = 30°$).

8 Conclusion

A prominent public model of orbital data-center radiator behavior is internally reproducible under its stated assumptions but, at its own default geometry, its Earth view factor is wrong by a large factor. Replacing the coded value with the exact tilted-plate-to-sphere view factor, and changing nothing else, raises the predicted edge-on equilibrium temperature by 6.35 K, from 335.75 K to 342.10 K. That correction to the code as executed separates into a +5.77 K model-form geometric error—replacing the intended 5% floor by the exact geometry—and a +0.58 K floating-point edge-case artifact in the orbit average; the correction is positive at every sampled β and monotone on the evaluated 0.5° grid, and largest at the default. Extending the static balance to an orbit-coupled transient model shows that a steady, averaged-load sizing does not under-predict the mean panel temperature—energy balance pins $\langle T^4 \rangle = T^4_{\mathrm{steady}}$, so the mean sits at or below steady—but under-predicts the orbital peak by up to several kelvin for a light panel. Both results are reproducible and machine verified, and both are offered as cross-model verification, not as validation against this proposed orbital-data-center radiator configuration, for which we found no public flight data. For a temperature architecture in which the rejection temperature enters the required area quartically, a several-kelvin error at the design point and a multi-kelvin unmodeled peak are not rounding; they are the kind of margin a flight design has to keep.

Acknowledgments

This work was produced through an author-directed, iterative, adversarial workflow across multiple AI systems: derivation and implementation with Claude (Anthropic), literature-armed review with Perplexity, and an eight-round software review with independent computer-algebra verification by GPT-5.5 with the Wolfram plugin, concluding without a remaining flagged defect. The author directed the work and takes responsibility for the result. The supporting software, test suite, frozen oracle, verification script, and figures are openly available in the project repository; the software-review reports are archived alongside them.

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